§1.3 Primes HW. 16, 41, 42, 43, 44, 48

Def. An integer $p>1$ is called a prime if there is no divisor d of p with $1<d<p$. (vs. composite)

Thm (Fundamental Theorem of Arithmetic)

Every integer $n>1$ can be uniquely expressed as a product of primes (w/ perhaps only one factor)

<u>pf of Idea</u>

i) Existence : If n is not a prime, $n=n_1n_2$, $0<n_i<n$.

Continue until it becomes prime.

ii) Uniqueness : Supp $n=\prod_{i=1}^{n} p_i^{e_i} = \prod_{j=1}^{m} q_j^{d_j}$, p_i, q_j dis. prime distinct primes, $e_i, d_j \in \mathbb{Z}^+$

Enough To Show that $n=m$ and each $p_i = q_j$ for some j.

If a prime p divides ab, p divides a or b. ///

Q. Does it hold for a set diff. from $\mathbb{Z}$?

$\boxed{\text{e.g. 1}}$ Let $\mathcal{C} = \mathbb{Z}[\sqrt{-6}] := \{a+b\sqrt{-6} : a, b \in \mathbb{Z}\}$

Is $\mathcal{C}$ a UFD (Unique Factorization Domain)?

Issue : prime $\Rightarrow$ divisor $\Rightarrow$ nontrivial (unit ??)

$ab=1 \Leftrightarrow a, b$ unit

$(\because)$ $p = 1\cdot 1\cdot 1\cdot p = a\cdot b\cdot p$

i) $\text{Norm}(a+b\sqrt{-6}) = (a+b\sqrt{-6})(a+b(-\sqrt{-6})) = a^2+6b^2$

$\hookrightarrow$ multiplicative $\overline{a+b\sqrt{-6}}$ 켤레무리수 conjugate

$N(ab) = N(a)N(b)$

ii) $a+b\sqrt{-6}$ is a unit if $\exists\, c+d\sqrt{-6} \in \mathcal{C}$

s.t. $(a+b\sqrt{-6})(c+d\sqrt{-6}) = 1$. identity 항등원

$\overset{\text{Norm}}{\Leftrightarrow}$ $(a^2+6b^2)(c^2+6d^2) = N(1) = 1$, $a, b, c, d \in \mathbb{Z}$.

$\Rightarrow a^2+6b^2 = 1 \quad \Rightarrow a^2 = 1 \quad \Rightarrow a = \pm 1,\ b = 0.$

$\therefore$ $\mathcal{O}$의 units $= \{\pm 1\}$

iii) 5 is a prime in $\mathcal{O}$

because $5 = (a+b\sqrt{-6})(c+d\sqrt{-6})$

$\overset{\text{Norm}}{\Longrightarrow} 25 = \overset{1}{5}(a^2+6b^2)\overset{25}{5}(c^2+6d^2)$, $a, b, c, d \in \mathbb{Z}$

$\Rightarrow a^2+6b^2 = 1, \cancel{5} \quad \Rightarrow a = \pm 1,\ b = 0$

iv) 2 is a prime in $\mathcal{O}$ similarly

v) $10 = 2 \cdot 5 = (2+\sqrt{-6})(2-\sqrt{-6})$ where $2 \pm \sqrt{-6}$ not a unit.

$\therefore$ Give two prime factorizations of 10.

Consequently, $\mathcal{O}$ is not a UFD.

$\boxed{\text{e.g.2}}$ How about $\mathbb{Z}[i]$? Yes $\longrightarrow$ Gaussian Integer

$\mathbb{Z}[\sqrt{d}]$ is UFD if $d = -163, -67, -43, -19, -11,$
$-7, -3, -2, -1, 1, 2, 3, 5 \cdots$

Thm 1.17 (Euclid) The number of primes is infinite

pf) H.W.

Thm 1.18 There are arbitrarily large gaps in the series of primes.

e.g. $\overbrace{2, 3}^{\text{run}}, 4, 5, 6, 7, \overbrace{8, 9, 10}^{\text{gap}}, 11, 12, 13, 14,$

pf) Consider $\underbrace{(k+1)!+2, \cdots, (k+1)!+(k+1)}_{k>4}$, each of which is composite. ///

Prime Number Theorem

Let $\pi(x) := \{ y \le x \mid y \text{ is prime} \}$

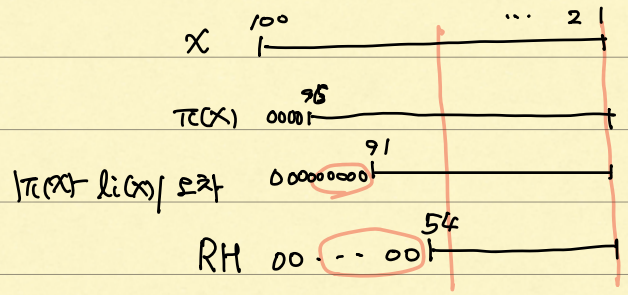
$$\lim_{x\to\infty} \frac{\pi(x)}{x/\ln x} = 1 \quad \left(\text{or } \pi(x) = \frac{x}{\ln x} + O\left(\frac{x}{\ln^2 x}\right)\right)$$

$\Rightarrow$ x가 소수일 확률 $\dfrac{x/\ln x}{x} = \dfrac{1}{\ln x}$

e.g. $x \simeq e^{100}$ $\quad \pi(x) = \dfrac{x}{100} + O\left(\dfrac{x}{100^2}\right)$ $\quad$ 확률 $\dfrac{1}{100}$

∘ Riemann Hypothesis. $\quad |\pi(x) - li(x)| < \dfrac{1}{8\pi}\sqrt{x}\log x$

$$li(x) = \int_0^x \frac{1}{\ln x}\,dx = \frac{x}{\ln x} + O\left(\frac{x}{\ln^2 x}\right)$$



$x = 12345\,12345\cdots$

$li(x) =$

소수의 개수 예측.

$\neq$

소수 여부